

EXPERIMENT – 8

Privilege Escalation Using TryHackMe Platform

Aim:

To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required:

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Procedure:

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using your registered credentials.

Step 2: Open Privilege Escalation Room

1. In the search bar, type **Linux Privilege Escalation**.
2. Click on the appropriate room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Initial Enumeration

1. Check current user:
`whoami`
2. Check system information:
`uname -a`
3. Check sudo permissions:
`sudo -l`

Step 5: Identify Privilege Escalation Vectors

Common techniques explored:

- Sudo misconfigurations
- SUID binaries
- Weak file permissions
- Environment variable abuse
- Cron job misconfigurations

Example command:

```
find / -perm -4000 2>/dev/null
```

Step 6: Exploit Vulnerability

1. Execute the vulnerable binary or misconfigured command.
2. Escalate privileges to root user.
3. Verify escalation:
whoami

Step 7: Capture Flags / Complete Tasks

1. Obtain root-level access.
2. Capture the required flags.
3. Mark all tasks as completed.

Output:

Task 1 Introduction

Privilege escalation is a journey. There are no silver bullets, and much depends on the specific configuration of the target system. The kernel version, installed applications, supported programming languages, other users' passwords are a few key elements that will affect your road to the root shell.

This room was designed to cover the main privilege escalation vectors and give you a better understanding of the process. This new skill will be an essential part of your arsenal whether you are participating in CTFs, taking certification exams, or working as a penetration tester.

Answer the questions below

Read the above.

No answer needed

✓ Correct Answer

Task 2 What is Privilege Escalation?

What does "privilege escalation" mean?

At its core, Privilege Escalation usually involves going from a lower permission account to a higher permission one. More technically, it's the exploitation of a vulnerability, design flaw, or configuration oversight in an operating system or application to gain unauthorized access to resources that are usually restricted from the users.



Why is it important?

It's rare when performing a real-world penetration test to be able to gain a foothold (initial access) that gives you direct administrative access. Privilege escalation is crucial because it lets you gain system administrator levels of access, which allows you to perform actions such as:

- Resetting passwords
- Bypassing access controls to compromise protected data
- Editing software configurations
- Enabling persistence
- Changing the privilege of existing (or new) users
- Execute any administrative command

Answer the questions below

Read the above.

No answer needed

✓ Correct Answer

Task 3 Enumeration

Note: Launch the target machine attached to this task to follow along.

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen
Password: Password1

▶ Start Machine

Answer the questions below

What is the hostname of the target system?

wade7363

✓ Correct Answer

What is the Linux kernel version of the target system?

3.13.0-24-generic

✓ Correct Answer

What Linux is this?

Ubuntu 14.04 LTS

✓ Correct Answer

What version of the Python language is installed on the system?

2.7.6

✓ Correct Answer

What vulnerability seem to affect the kernel of the target system? (Enter a CVE number)

CVE-2015-1328

✓ Correct Answer

Task 4 ✓ Automated Enumeration Tools

Several tools can help you save time during the enumeration process. These tools should only be used to save time knowing they may miss some privilege escalation vectors. Below is a list of popular Linux enumeration tools with links to their respective Github repositories.

The target system's environment will influence the tool you will be able to use. For example, you will not be able to run a tool written in Python if it is not installed on the target system. This is why it would be better to be familiar with a few rather than having a single go-to tool.

- **LinPeas**: <https://github.com/carlospolop/privilege-escalation-awesome-scripts-suite/tree/master/linPEAS>
- **LinEnum**: <https://github.com/rebootuser/LinEnum>
- **LES (Linux Exploit Suggester)**: <https://github.com/mzet-/linux-exploit-suggester>
- **Linux Smart Enumeration**: <https://github.com/diego-treitos/linux-smart-enumeration>
- **Linux Priv Checker**: <https://github.com/linted/linuxprivchecker>

Answer the questions below

Install and try a few automated enumeration tools on your local Linux distribution

No answer needed

✓ Correct Answer

Task 5 ✓ Privilege Escalation: Kernel Exploits

Note: Launch the target machine attached to this task to follow along.

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen

Password: Password1

Answer the questions below

find and use the appropriate kernel exploit to gain root privileges on the target system.

No answer needed

✓ Correct Answer



What is the content of the flag1.txt file?

THM-28392872729920

✓ Correct Answer

Task 6 ✓ Privilege Escalation: Sudo

Note: Launch the target machine attached to this task to follow along.

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen

Password: Password1

Answer the questions below

How many programs can the user "karen" run on the target system with sudo rights?

3

✓ Correct Answer

What is the content of the flag2.txt file?

THM-402028394

✓ Correct Answer

How would you use Nmap to spawn a root shell if your user had sudo rights on nmap?

sudo nmap --interactive

✓ Correct Answer

What is the hash of frank's password?

\$6\$2.sUUDsOLIpXKxcr\$elmtgFExyr2ls4jsghdD3DHLHHP9X50lv.jNmwo/BJpphrPRJWjelWEz2HH.joV14aDEwW1c3CahzB1uaq

✓ Correct Answer

Task 7 Privilege Escalation: SUID



Note: Launch the target machine attached to this task to follow along.

▶ Start Machine

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen

Password: Password1

Answer the questions below

Which user shares the name of a great comic book writer?

gerryconway

✓ Correct Answer

What is the password of user2?

Password1

✓ Correct Answer

What is the content of the flag3.txt file?

THM-3847834

✓ Correct Answer

Task 8 Privilege Escalation: Capabilities



Note: Launch the target machine attached to this task to follow along.

▶ Start Machine

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen

Password: Password1

Answer the questions below

Complete the task described above on the target system

No answer needed

✓ Correct Answer

How many binaries have set capabilities?

6

✓ Correct Answer

What other binary can be used through its capabilities?

view

✓ Correct Answer

What is the content of the flag4.txt file?

THM-9349843

✓ Correct Answer

Task 9 Privilege Escalation: Cron Jobs



Note: Launch the target machine attached to this task to follow along.

Start Machine

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen

Password: Password1

Answer the questions below

How many user-defined cron jobs can you see on the target system?

4

✓ Correct Answer

What is the content of the flag5.txt file?

THM-383000283

✓ Correct Answer

What is Matt's password?

123456

✓ Correct Answer

Task 10 Privilege Escalation: PATH



Note: Launch the target machine attached to this task to follow along.

Start Machine

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen

Password: Password1

Answer the questions below

What is the odd folder you have write access for?

/home/murdoch

✓ Correct Answer



Exploit the \$PATH vulnerability to read the content of the flag6.txt file.

No answer needed

✓ Correct Answer



What is the content of the flag6.txt file?

THM-736628929

✓ Correct Answer

Task 11 Privilege Escalation: NFS



Note: Launch the target machine attached to this task to follow along.

Start Machine

You can launch the target machine and access it directly from your browser.

Alternatively, you can access it over SSH with the low-privilege user credentials below:

Username: karen

Password: Password1

Answer the questions below

How many mountable shares can you identify on the target system?

3

✓ Correct Answer

How many shares have the "no_root_squash" option enabled?

3

✓ Correct Answer

Gain a root shell on the target system

No answer needed

✓ Correct Answer

What is the content of the flag7.txt file?

THM-89384012

✓ Correct Answer

By now you have a fairly good understanding of the main privilege escalation vectors on Linux and this challenge should be fairly easy.

[▶ Start Machine](#)

You have gained SSH access to a large scientific facility. Try to elevate your privileges until you are Root.

We designed this room to help you build a thorough methodology for Linux privilege escalation that will be very useful in exams such as OSCP and your penetration testing engagements.

Leave no privilege escalation vector unexplored, privilege escalation is often more an art than a science.

You can access the target machine over your browser or use the SSH credentials below.

- Username: leonard
- Password: Penny123

Answer the questions below

What is the content of the flag1.txt file?

THM-42828719920544

✓ Correct Answer

What is the content of the flag2.txt file?

THM-168824782390238

✓ Correct Answer

Result:

Thus, privilege escalation was successfully performed using the TryHackMe platform. The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.